CUSTOMER SHOPPING BEHAVIOR ANALYSIS

END-TO-END DATA ANALYTICS PROJECT REPORT

Project Domain	Retail & Consumer Analytics
Tools Used	Python (pandas) · PostgreSQL · Power BI
Dataset	Customer Retail Transactions — 8,000 records × 19 columns
Analysis Type	EDA · SQL Business Intelligence · RFM Segmentation · Cohort Analysis · CLV
Dashboard Pages	3 Pages — Executive Overview · Behavior Drivers · Advanced Analytics
Total Revenue Analysed	₹21,71,90,250 (₹217M+)
Date Completed	March 2026

1. Project Overview

This project delivers a complete, industry-grade end-to-end data analytics workflow for a leading retail company. The goal is to translate 8,000 customer retail transactions into strategic business intelligence using Python for data preparation, PostgreSQL for structured querying, and Power BI for interactive visualisation.

1.1 Business Problem Statement

A leading retail company has observed shifting purchase patterns across demographics, product categories, and sales channels. Management wants to understand which factors — discounts, ratings, seasons, payment preferences — drive consumer decisions and repeat purchases, to improve customer engagement and optimise marketing strategy.

Core Business Question

"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimise marketing and product strategies?"

1.2 Project Deliverables

- **Data Preparation & EDA (Python):** Clean and transform the raw dataset; load into PostgreSQL.
- **Data Analysis (SQL):** 16 business queries + advanced RFM, Cohort Retention, and CLV analysis.
- **Visualisation (Power BI):** 3-page interactive dashboard with 5 cross-page slicers.
- **Project Report:** This document is a comprehensive documentation of all findings.

2. Dataset Summary

The dataset contains 8,000 customer retail transactions across 4 product categories. After Python cleaning, 2 new feature columns were engineered, bringing the final dataset to 21 columns.

8,000	**19**	**21**	**4**
Total Records	Raw Columns	Cleaned Columns	Product Categories
8	**10**	**2024–2025**	**₹217M+**
Unique Locations	Unique Products	Date Range	Total Revenue

2.1 Raw Dataset — Missing Values

Column	Missing Count	% Missing	Treatment
Size	5,479	68.5%	Imputed/handled in cleaning
Promo Code Used	150	1.9%	Imputed using mode — column retained (distinct from discount _applied)
Review Rating	120	1.5%	Imputed using the median per product category
Shipping Type	100	1.25%	Imputed using the mode per category

2.3 Key Dataset Features

Feature Group	Columns	Description
Customer Demographics	age, gender, location	Customer profile attributes
Purchase Details	item_purchased, category, purchase_amount, season, size, color	Transaction-level data (INR)
Dates	purchase_date	Transaction date — used for Cohort & RFM analysis
Subscription & Loyalty	subscription_status, previous_purchases, frequency_of_purchases	Retention indicators
Engagement Signals	review_rating, discount_applied, promo_code_used, shipping_type, payment_method	Behavioral triggers
Engineered (Python)	age_group, purchase_frequency_days	Created during EDA — not in raw data

3. Data Preparation & EDA (Python)

Python (Jupyter Notebook) was used for all data ingestion, cleaning, feature engineering, and exploratory analysis. The cleaned DataFrame was then loaded into PostgreSQL via SQLAlchemy for SQL-based analysis.

3.1 Data Loading & Initial Exploration

- Imported 8,000-record CSV using pandas.
- **df.info()** used to inspect column types, null counts, and memory — revealed 19 columns with mixed types.
- **df.describe()** for statistical summary: Purchase Amount ₹1,245 – ₹1,24,251; Age 18 – 69; Review Rating 2.5 – 5.0.
- Identified 4 columns with missing values (Size: 5,479 | Promo Code: 150 | Review Rating: 120 | Shipping Type: 100).

3.2 Data Cleaning Steps

Step	Action Taken	Result
Missing — Review Rating	Imputed using median rating per product category	120 nulls resolved
Missing — Shipping Type	Imputed using the mode per category	100 nulls resolved
Missing — Size	Handled appropriately (high missing rate: 68.5%)	5,479 nulls resolved
Missing — Promo Code Used	Filled then verified against Discount Applied	150 nulls resolved
Column Naming	Renamed all 19 columns to snake_case	SQL-compatible naming

Step	Action Taken	Result
Redundancy Check	Verified discount_applied vs promo_code_used — found to be DISTINCT (3,721 rows differ)	Both columns retained for independent analysis
Data Types	Verified purchase_date as datetime, amounts as numeric	No type mismatches
Duplicates	Checked for duplicate customer_id entries	No duplicates found

3.3 Feature Engineering

- age_group: Customer ages binned into 4 groups: Young Adult (18–30): 2,117 | Middle-aged (31–50): 2,098 | Adult (51–60): 1,923 | Senior (61+): 1,862
- purchase_frequency_days: Categorical frequency mapped to numeric days — Monthly: 30 days (2,787 customers) | Quarterly: 90 days (1,642) | Bi-Weekly: 14 days (1,570) | Weekly: 7 days (1,239) | Annually: 365 days (762)

3.4 Database Integration

- Connected Python script to PostgreSQL using SQLAlchemy engine.
- Loaded cleaned 8,000-row DataFrame into schema: customer_behavior.customer.
- Database used as the single source of truth for all downstream SQL analysis.

4. Data Analysis (SQL — PostgreSQL)

16 structured business queries were written and executed in PostgreSQL to answer critical business questions across revenue, customer behaviour, loyalty, and engagement. Additionally, three advanced analytical models were built: RFM Segmentation, Cohort Retention Analysis, and Customer Lifetime Value.

4.1 Business Intelligence Queries — Results

Q.1

```
--Q1. What is the total revenue generated by male vs. female customers?
SELECT gender, SUM(purchase_amount) as revenue
FROM customer_behavior.customer
GROUP BY gender;
```

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	gender text	revenue numeric
1	Female	114229912
2	Male	102960338

Q.3

```
-- Q3. Which are the top 5 products with the highest average review rating?
SELECT item_purchased, ROUND(AVG(review_rating::numeric),2) as "Average Product Rating"
FROM customer_behavior.customer
GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC
limit 5;
```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	item_purchased text	Average Product Rating numeric
1	Headphones	4.00
2	Boots	4.00
3	Backpack	3.99
4	Sneakers	3.98
5	Laptop	3.98

Q.6

```sql
--Q6. What is the revenue contribution of each age group?
SELECT
    age_group,
    SUM(purchase_amount) AS total_revenue
FROM customer_behavior.customer
GROUP BY age_group
ORDER BY total_revenue desc;
```

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	age_group text	total_revenue numeric
1	Middle-aged	57359308
2	Young Adult	57211402
3	Adult	52055193
4	Senior	50564347

Q.11

```sql
--Q.11 Can we segment customers based on behavioral patterns?
SELECT
    CASE
        WHEN previous_purchases >= 20 THEN 'Highly Loyal'
        WHEN previous_purchases BETWEEN 10 AND 19 THEN 'Moderately Loyal'
        ELSE 'Low Engagement'
    END AS customer_segment,
    COUNT(*) AS total_customers,
    ROUND(AVG(purchase_amount),2) AS avg_spend,
    ROUND(AVG(review_rating::numeric),0) AS avg_rating
FROM customer_behavior.customer
GROUP BY customer_segment
ORDER BY avg_spend DESC;
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	customer_segment text	total_customers bigint	avg_spend numeric	avg_rating numeric
1	Low Engagement	3689	27410.29	4
2	Moderately Loyal	4270	27039.51	4
3	Highly Loyal	41	14998.71	4

Q.12

```sql
--Q.12 Do promo codes create long-term customers or one-time buyers?
SELECT
    promo_code_used,
    COUNT(*) AS total_orders,
    ROUND(AVG(previous_purchases),0) AS avg_repeat_purchases,
    ROUND(AVG(purchase_amount),2) AS avg_spend,
    ROUND(AVG(review_rating:: numeric),2) AS avg_rating
FROM customer_behavior.customer
GROUP BY promo_code_used;
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	promo_code_used text	total_orders bigint	avg_repeat_purchases numeric	avg_spend numeric	avg_rating numeric
1	No	5485	10	27237.83	3.98
2	Yes	2365	10	26887.51	3.97
3	No Promo	150	10	28011.95	3.96

	Business Question	Actual Result / Key Finding
Q2	Discount users who spent above average	Customers who used discounts but still spent above avg purchase amount of ₹27,149
Q4	Subscriber vs. Non-subscriber spend	Subscribers: ₹27,319 avg (2,763 customers) | Non-subscribers: ₹27,058 avg (5,237 customers) Negligible spend difference despite subscription
Q5	Top 5 discount-dependent products	Jeans: 42.97% | Boots: 42.01% | Jacket: 41.30% | Headphones: 41.14% | Watch: 40.28%
Q7	Factors driving repeat purchases	Subscribed customers who also use discounts show highest avg repeat purchase behavior
Q8	Rating vs. spending correlation	Review ratings are consistent (avg 3.97) across spend levels — no strong correlation
Q9	Seasonal customer behavior	Winter: ₹27,730 avg spend (highest) | Summer: ₹26,379 (lowest) | All seasons within 5% of each other
Q10	Subscriber behavioral profile	2,763 subscribers vs 5,237 non-subscribers | Similar satisfaction (avg rating ~3.97 both groups)
Q13	Category that builds most loyal customers	Clothing: avg loyalty 10.05 | Footwear: 10.03 | Electronics: 9.99 | Accessories: 9.97 — all nearly equal
Q14	Payment preference & customer maturity	Cash users: avg age 44.07 (oldest) | PayPal: 43.79 | Credit Card: 43.62 | Google Pay: 42.85 (youngest)
Q15	High spenders vs. engagement	Low Spenders (<₹20K): 5,465 customers | High Spenders (₹50K+): 1,581 | Medium: 954
Q16	Purchase frequency vs. actual engagement	Monthly buyers (2,787): most frequent segment | Annual buyers (762): least frequent

4.2 Advanced SQL Analysis —

- RFM Segmentation

RFM (Recency, Frequency, Monetary) analysis was performed using PostgreSQL CTEs and NTILE(4) window functions to score each customer across three dimensions and classify them into actionable segments.

```sql
--RFM Analysis
WITH rfm_base AS (
    SELECT
        customer_id,
        MAX(purchase_date::date) AS last_purchase_date,
        COUNT(*) AS frequency,
        SUM(purchase_amount) AS monetary
    FROM customer_behavior.customer
    GROUP BY customer_id
),

```

```sql
rfm_scores AS (
    SELECT *,
        NTILE(4) OVER (ORDER BY last_purchase_date DESC) AS recency_score,
        NTILE(4) OVER (ORDER BY frequency DESC) AS frequency_score,
        NTILE(4) OVER (ORDER BY monetary DESC) AS monetary_score
    FROM rfm_base
),

rfm_final AS (
    SELECT *,
           (recency_score + frequency_score + monetary_score) AS rfm_total_score
    FROM rfm_scores
```

```sql

SELECT
    CASE
        WHEN rfm_total_score >= 10 THEN 'Champions'
        WHEN rfm_total_score BETWEEN 7 AND 9 THEN 'Loyal Customers'
        WHEN rfm_total_score BETWEEN 4 AND 6 THEN 'Potential'
        ELSE 'At Risk'
    END AS segment,
    COUNT(*) AS total_customers,
    ROUND(AVG(monetary)::numeric, 2) AS avg_revenue
FROM rfm_final
GROUP BY segment
ORDER BY avg_revenue DESC;
```

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	segment text	total_customers bigint	avg_revenue numeric
1	At Risk	500	76959.09
2	Potential	2517	53495.73
3	Loyal Custome...	2987	10999.82
4	Champions	1996	5613.98

- Cohort Retention Analysis

Customers were grouped by their first purchase month (cohort month). Monthly activity was tracked across subsequent months using DATE_TRUNC and a self-JOIN to measure retention over time. This identifies which acquisition cohorts retain customers most effectively and at what point drop-off occurs.

```sql
--Cohort Retention Analysis
WITH first_purchase AS (
    SELECT
        customer_id,
        MIN(DATE_TRUNC('month', purchase_date::timestamp)) AS cohort_month
    FROM customer_behavior.customer
    GROUP BY customer_id
),
```

```sql
),

monthly_activity AS (
    SELECT
        c.customer_id,
        DATE_TRUNC('month', c.purchase_date::timestamp) AS activity_month,
        f.cohort_month
    FROM customer_behavior.customer c
    JOIN first_purchase f
        ON c.customer_id = f.customer_id
)

SELECT
    cohort_month,
    activity_month,
    COUNT(DISTINCT customer_id) AS active_customers
FROM monthly_activity
GROUP BY cohort_month, activity_month
ORDER BY cohort_month, activity_month;
```

Data Output Messages Notifications

Showing r

	cohort_month timestamp without time zone	activity_month timestamp without time zone	active_customers bigint
1	2024-01-01 00:00:00	2024-01-01 00:00:00	346
2	2024-02-01 00:00:00	2024-02-01 00:00:00	331
3	2024-03-01 00:00:00	2024-03-01 00:00:00	361
4	2024-04-01 00:00:00	2024-04-01 00:00:00	341
5	2024-05-01 00:00:00	2024-05-01 00:00:00	322
6	2024-06-01 00:00:00	2024-06-01 00:00:00	326
7	2024-07-01 00:00:00	2024-07-01 00:00:00	311
8	2024-08-01 00:00:00	2024-08-01 00:00:00	330
9	2024-09-01 00:00:00	2024-09-01 00:00:00	305
10	2024-10-01 00:00:00	2024-10-01 00:00:00	352
11	2024-11-01 00:00:00	2024-11-01 00:00:00	331
12	2024-12-01 00:00:00	2024-12-01 00:00:00	348
13	2025-01-01 00:00:00	2025-01-01 00:00:00	320
14	2025-02-01 00:00:00	2025-02-01 00:00:00	302
15	2025-03-01 00:00:00	2025-03-01 00:00:00	322
16	2025-04-01 00:00:00	2025-04-01 00:00:00	343
17	2025-05-01 00:00:00	2025-05-01 00:00:00	332
18	2025-06-01 00:00:00	2025-06-01 00:00:00	339
19	2025-07-01 00:00:00	2025-07-01 00:00:00	366
20	2025-08-01 00:00:00	2025-08-01 00:00:00	348
21	2025-09-01 00:00:00	2025-09-01 00:00:00	323
22	2025-10-01 00:00:00	2025-10-01 00:00:00	347
23	2025-11-01 00:00:00	2025-11-01 00:00:00	337
24	2025-12-01 00:00:00	2025-12-01 00:00:00	317

- Customer Lifetime Value

Top 20 highest-value customers were identified by ranking on total lifetime spend (SUM of purchase_amount per customer_id). For each customer, the query returned: total orders placed, cumulative lifetime value, and average order value — enabling targeted VIP customer relationship management.

```sql
--Customer Lifetime Value
SELECT
    customer_id,
    COUNT(*) AS total_orders,
    SUM(purchase_amount) AS lifetime_value,
    ROUND(AVG(purchase_amount),2) AS avg_order_value
FROM customer_behavior.customer
GROUP BY customer_id
ORDER BY lifetime_value DESC
LIMIT 20;
```

Data Output Messages Notifications

Showing rows: 1 to 20 Page No: 1 of 1

	customer_id bigint	total_orders bigint	lifetime_value numeric	avg_order_value numeric
1	10283	1	124251	124251.00
2	14211	1	124251	124251.00
3	12466	1	124168	124168.00
4	8668	1	124085	124085.00
5	8099	1	124085	124085.00
6	14679	1	124085	124085.00
7	12166	1	124085	124085.00
8	8362	1	124085	124085.00
9	11568	1	123919	123919.00
10	11333	1	123919	123919.00
11	9538	1	123836	123836.00
12	13327	1	123836	123836.00
13	14536	1	123836	123836.00
14	15019	1	123753	123753.00
15	10520	1	123753	123753.00
16	13083	1	123504	123504.00
17	11196	1	123504	123504.00
18	15203	1	123504	123504.00
19	13830	1	123504	123504.00
20	9092	1	123421	123421.00

5. Power BI Dashboard

A 3-page interactive Power BI dashboard was built to present insights at multiple levels — from executive KPIs to deep behavioural and advanced analytics. All 3 pages share 5 cross-page slicers: Year, Gender, Age Group, Category, and Location.

Page 1 — Executive Overview

₹217M	8,000	₹27,149	68 Days
Total Revenue	Total Customers	Avg. Order Value	Avg. Purchase Interval

Visual	Type	Key Insight
Monthly Revenue Trend (2024 vs 2025)	Line Chart	Consistent monthly revenue ~₹8–9M across both years with no sharp seasonal spikes
Revenue by Product Category	Bar Chart	Electronics dominates at ₹163M (75% of total) | Accessories ₹20M | Clothing ₹17M | Footwear ₹15M
Customer Segment Distribution	Donut Chart	Potential Loyalists: 41.31% | Champions: 39.39% | Loyal Customers: 19.3%
Revenue by Customer Segment	Bar Chart	Champions generate ₹179M despite being 39% of customers — power law distribution
Avg. Previous Purchases KPI	Card	10 average previous purchases across all 8,000 customers

Page 2 — Behavior Drivers

₹27K	68 Days	40.2%	3.97 / 5
Avg Purchase Amount	Avg Purchase Interval	Discount Usage Rate	Avg Customer Rating

Visual	Type	Key Insight
Purchase Frequency by Age Group	Bar Chart	Young Adults & Middle-aged customers show the highest purchase frequency
Rating Impact on Revenue	Bar Chart	Products rated 3.5–4.5 contribute the highest revenue — quality matters
Seasonal Customer Behavior	Line Chart	Winter: highest avg spend ₹27,730 | Summer: lowest ₹26,379 | All within 5% range
Shipping Preference by Segment	Matrix Table	Standard shipping dominates all segments | Champions use more Express shipping
Payment Behavior Analysis	Pie Chart	Credit Card: 40.7% | Debit Card: 24.8% | PayPal: 14.7% | Apple Pay: 8.1% | Google Pay: 6.9%
Discount Impact on Revenue	Bar Chart	Non-discounted: ₹131.8M | Discounted: ₹85.4M — discounts reduce revenue contribution

Page 3 — Advanced Analytics

42%	434	68 Days	₹27K
Repeat Purchase Rate	Avg. Recency (days)	Avg. Purchase Interval	Avg. Customer Value

Visual	Type	Key Insight
Customer Spending by Recency	Line Chart	Customers with 0 – 30-day recency have the highest average. spend — recency drives purchase value
Customer Purchase Interval Distribution	Clustered Bar	Champions dominate 0 – 30-day intervals | At Risk spread across longer intervals
Customer Engagement Analysis	Scatter Plot	Champions clustered at high recency + high frequency | At Risk scattered at low values

6. Key Insights & Findings

6.1 Revenue & Customer Segmentation

- Female customers (4,180) contribute slightly higher total revenue ₹114.2M vs Male (3,820) ₹102.9M — contrary to typical retail assumptions
- Champions (39.39% of customers) generate ₹179M — approximately 82% of total revenue — a classic Pareto distribution
- Potential Loyalists are the largest segment (41.31%) and represent the biggest growth opportunity for the business
- Only 41 customers qualify as Highly Loyal (20+ purchases) — loyalty program needs strengthening

6.2 Product & Category Performance

- Electronics dominates revenue at ₹163M (75% of total), far outpacing Clothing (₹17M), Accessories (₹20M), and Footwear (₹15M)
- Top-rated products (avg 4.0): Headphones and Boots — both premium products with quality perception
- Jeans (42.97%), Boots (42.01%), and Jacket (41.30%) are the most discount-dependent products
- All 4 categories show nearly identical average loyalty scores (≈10) — no single category builds disproportionate loyalty.

6.3 Discount & Pricing Strategy

- 40.2% of all 8,000 purchases used a discount — generating ₹85.4M vs ₹131.8M from non-discounted purchases
- Promo codes show near-zero impact on loyalty: promo users avg 10.04 repeat purchases vs 10.01 for non-users
- Discounts attract transaction volume but compress margins — discount strategy needs to be product-specific

6.4 Subscription & Loyalty Dynamics

- Subscribers (2,763) avg spend ₹27,319 vs Non - subscribers (5,237) avg ₹27,058 — negligible difference of ₹261
- Subscription program is not differentiating customer value — exclusive subscriber benefits are needed
- Moderately Loyal customers (10–19 purchases) form the largest loyalty group at 4,270 — these are retention targets

6.5 Behavioral & Seasonal Patterns

- All 4 seasons show consistent avg spend within a 5% range — no significant seasonal demand spikes.
- Winter shows the highest avg spend ₹27,730 while Summer is slightly lower at ₹26,379
- 0–30 day recency customers have the highest average spend — recency is the strongest behavioural predictor.
- Cash users are the oldest average age of 44.07, while Google Pay users are the youngest (42.85) — generational payment pattern.

7. Business Recommendations

	Recommendation	Based On	Expected Impact
1	Launch VIP loyalty program for the Champions segment	39% of customers = 82% of ₹217M revenue	Protect and grow the highest-value customer base
2	Design targeted win-back campaigns for At Risk RFM customers	At-risk customers have the highest drop-off risk	Recover lapsing revenue before full churn
3	Unlock growth from Potential Loyalists (41.31% of base)	Largest segment — currently under-engaged	Convert to Champions to significantly boost revenue
4	Redesign subscription program with exclusive benefits	Subscribers & non-subscribers spend nearly identical amounts	Justify subscription cost and improve conversion rate
5	Reduce discount dependency on Jeans, Boots, Jacket	42–43% discount rate on these items compresses margins	Improve gross margin without losing volume
6	Use promo codes for new customer acquisition only — not retention	Promo codes show zero loyalty lift (10.01 vs 10.04 repeat)	Optimise promotion ROI, shift budget to loyalty programs
7	Target Electronics + Young Adults as primary revenue lever	Electronics: 75% revenue | Young Adults: ₹57M — both top-ranked	Maximise returns from the highest-revenue combination
8	Launch winter season campaigns to capitalise on the highest spend period	Winter avg spend ₹27,730 — highest of all seasons	Amplify natural seasonal behavior for revenue growth

8. Tools & Technologies Used

Tool / Library	Category	Purpose of This Project
Python 3.x	Programming	Data cleaning, EDA, feature engineering, and database connection
pandas	Python Library	Data manipulation, null handling, column transformation
matplotlib & seaborn	Python Library	Exploratory visualisations and distribution analysis during EDA
SQLAlchemy	Python Library	PostgreSQL database connection and DataFrame upload
PostgreSQL	Database	Structured data storage and all SQL-based business analysis
SQL (PostgreSQL dialect)	Query Language	16 business queries + RFM, Cohort Retention, and CLV analysis
Power BI Desktop	Visualization	3-page interactive dashboard with DAX measures and slicers
DAX	Power BI Language	Custom KPI measures, calculated columns, RFM segment logic
Jupyter Notebook	IDE	Python development, documentation, and EDA presentation

9. Conclusion

This end-to-end retail analytics project demonstrates a complete, industry-grade data analytics pipeline — from raw CSV ingestion through Python-based cleaning and feature engineering, to PostgreSQL business intelligence queries, Power BI interactive dashboards, and stakeholder-ready reporting.

The analysis of 8,000 customer transactions (totalling ₹217M+ in revenue) reveals that a small Champions segment drives the majority of business value, discount strategies are compressing margins without building loyalty, and the subscription program requires a significantly stronger value proposition. Seasonal demand is stable year-round, with recency emerging as the most reliable predictor of future customer spend.

The RFM model, Cohort Retention analysis, and CLV rankings together provide a data foundation for a targeted CRM strategy — enabling the business to reward its best customers, re-engage those at risk, and reallocate promotional spend toward proven retention levers.

9.1 Skills Demonstrated

- End-to-end data pipeline: Python → PostgreSQL → Power BI
- Advanced SQL: Window functions (NTILE, DATE_TRUNC), multi-level CTEs, subqueries
- Feature engineering: age binning, frequency-to-numeric conversion
- RFM modeling, Cohort Retention Analysis, and Customer Lifetime Value calculation
- Dashboard design: 3-page layered storytelling from KPIs to granular behavioral insights
- Business thinking: Translating data findings into concrete, actionable recommendations

9.2 Future Enhancements

- Predictive churn model using logistic regression or Random Forest in Python (scikit-learn)
- Time-series forecasting for seasonal revenue planning using SARIMA or Prophet
- Real-time pipeline integration using Apache Kafka or AWS for live Power BI refresh
- Full Customer Data Platform (CDP) with automated RFM segment triggers via email marketing